CONTACT INFORMATION

Department of Mathematics
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RESEARCH INTERESTS

Optimal transport, any-dimensional optimization, symmetry, manifold learning, high-dimensional probability.

EDUCATION

The Ohio State University, Columbus, Ohio, United States — 2022 – now

Ph.D. Candidate in Mathematics (Applied Track), GPA: 4.00/4.00
Thesis Advisor: **Dustin G. Mixon**

The Chinese University of Hong Kong, Shenzhen, China — 2018 – 2022

B.Sc. in Mathematics and Applied Mathematics (First Class Honours)

PUBLICATIONS

1. On the clustering behavior of sliding windows
(with Boris Alexeev, Dustin G. Mixon, Yan X. Zhang)
SIAM Journal on Mathematics of Data Science, 2026, [Accepted Manuscript][arXiv]

PREPRINTS

1. BalLOT: Balanced k-means clustering with optimal transport
(with Dustin G. Mixon)
Revised manuscript under review at *Mathematical Programming*, 2026, [Latest Draft][arXiv]

2. Wasserstein bounds for ambient and latent smoothing on data manifolds
(with Wonjun Lee)
Manuscript submitted to *Journal of Machine Learning Research*, 2026, [Latest Draft][arXiv]

IN PROGRESS

1. Solving any-dimensional permutation-equivariant semidefinite programming
(with Soledad Villar and Dustin G. Mixon)

2. Intrinsic dimension estimation via Gaussian smoothing in Wasserstein distance
(with Wonjun Lee)

AWARDS & FELLOWSHIPS

Special Graduate Assignment
The Ohio State University, Department of Mathematics — Spring 2027

Undergraduate Research Award
The Chinese University of Hong Kong, Shenzhen — 2021

Academic Performance Award (Top 5%)
The Chinese University of Hong Kong, Shenzhen — 2020 - 2021

ACADEMIC ACTIVITIES

Summer School in High-Dimensional Probability and Its Applications
Northwestern University, Evanston — August 2026

SLMath Workshop on Statistical Optimal Transport
University of California, Berkeley — June 2025

MREG Workshop on Multiclass Perceptron
University of Michigan, Ann Arbor — August 2023

Teaching

The Ohio State University

Recitation Instructor

Math 1148 College Algebra (Instructor: Amit Vutha)	Spring 2026, Autumn 2026
Math 1156 Calculus for Biology (Instructor: Trent Ohl)	Autumn 2023
Math 2153 Calculus III (Instructor: Amit Vutha)	Spring 2023
Math 1151 Calculus I (Instructor: Prerona Dutta)	Autumn 2022

Grader

Math 3345 Foundations of Higher Mathematics (Instructor: Weichen Gu)	Autumn 2025
Math 2568H Honors Linear Algebra (Instructor: Dustin Mixon)	Autumn 2025
Math 5522H Honors Complex Analysis (Instructor: Jennifer Park)	Spring 2025
Math 5530H Honors Probability (Instructor: Neil Falkner)	Spring 2025
Math 6211 Real Analysis I, PhD–Level (Instructor: David Penneys)	Spring 2024
Math 2568 Linear Algebra (Instructor: Dustin Mixon)	Spring 2024

The Chinese University of Hong Kong, Shenzhen

Undergraduate Student Teaching Fellow

Real Analysis (Instructor: Wei-Ming Ni)	Spring 2021

Mentoring Program

The Ohio State University

Directed Reading Program

Probability, Concentrations, and Programming
Mentee: Avery Stevens — Fall 2023

Interdisciplinary Experience

Multi-omics integration platform for prognostic biomarker discovery [GitHub Link]
Data Science Bootcamp, Erdős Institute — November 2025

Programming Languages

Python, MATLAB, R, LaTeX

Languages

English (fluent), Chinese (native in Cantonese and Mandarin)